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Lecture 7: Tuesday 30 September2024

Consumer Preferences

Business Economics, Climate Change and the Environment

Simon Quinn

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Module outline

Lecture	Date	Topic
1	9 September	Growth, Transformation and Transition
2	11 September	Introduction to Game Theory
3	16 September	Game Theory and Auctions
4	17 September	Game Theory and the Environment
5	23 September	Academic Papers Explained
6	25 September	Academic Papers Explained
7	30 September	Consumer Preferences
8	2 October	Institutions, Power and Inequality
9	7 October	Supply and Demand
10	9 October	Market Successes and Failures

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Today's objectives

Today, we will:

1. Discuss a graphical model of **consumer preferences**;

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1. Discuss a graphical model of **consumer preferences**;
2. Use that framework to think about **household consumption** subject to household budget constraints; and
3. Use the same framework to think about **government decision-making** subject to technological constraints.

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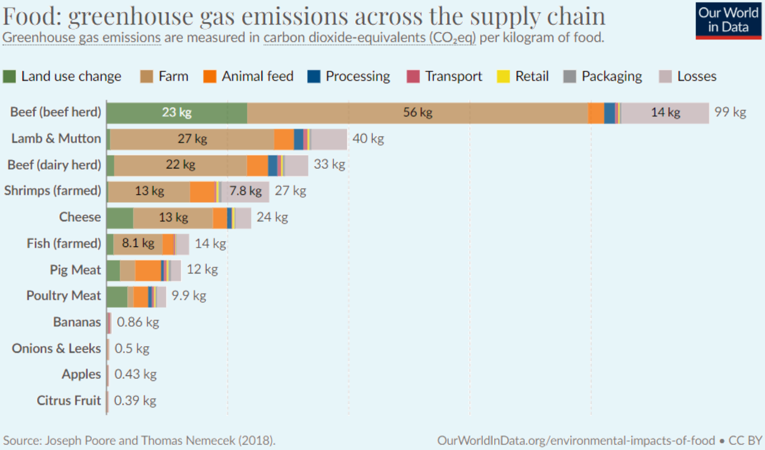
Greenhouse emissions and food choice



Using a graphical framework to compare 'bundles'

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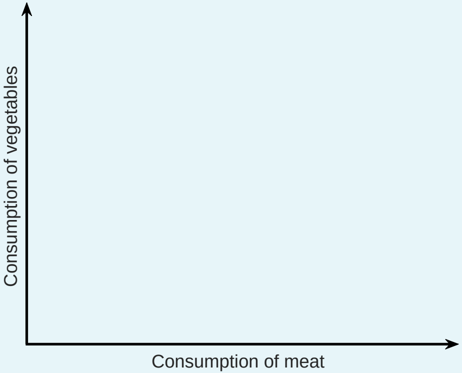

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- First, we start with a simple graph...
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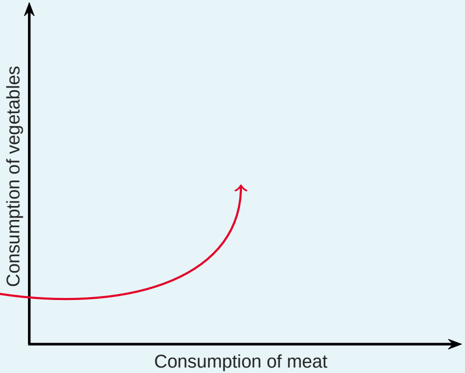

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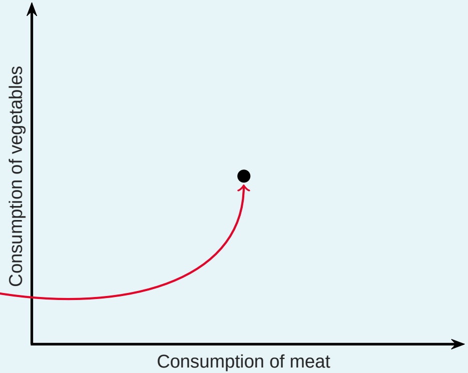

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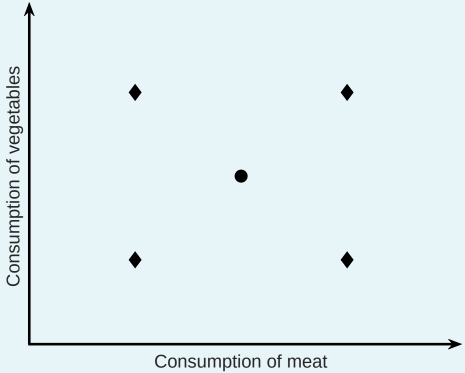
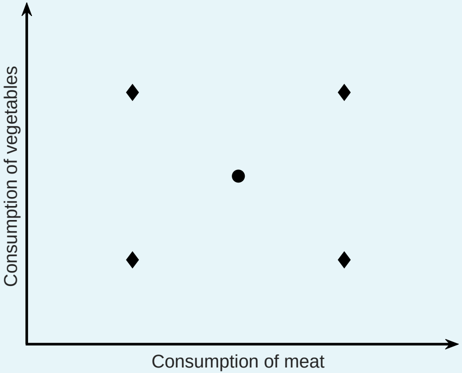


Using a graphical framework to compare 'bundles'

- First, we start with a simple graph...
- ...so any bundle is a **point**.
- Alternative choices are, therefore, just **different points**.
- To compare bundles, we need to make **assumptions about consumer behaviour**.



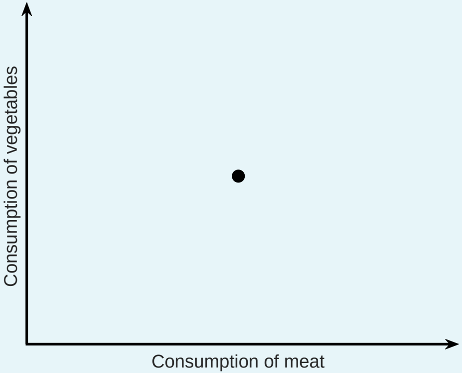
Introducing ‘non-satiation’

Definition: Non-satiation

‘Non-satiation’ means that the consumer will always **prefer more of each good to less**.

What does **non-satiation** allow us to say here?

- Let's compare the alternative options to our original bundle. . .



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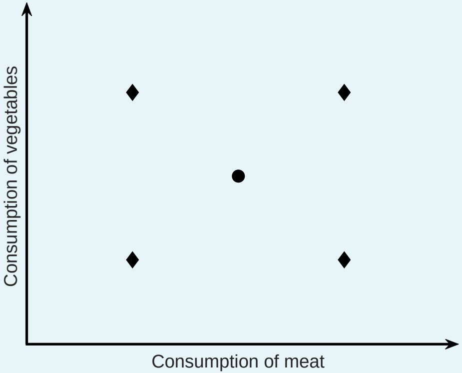

What does **non-satiation** allow us to say here?

- Let's compare the alternative options to our original bundle. . .
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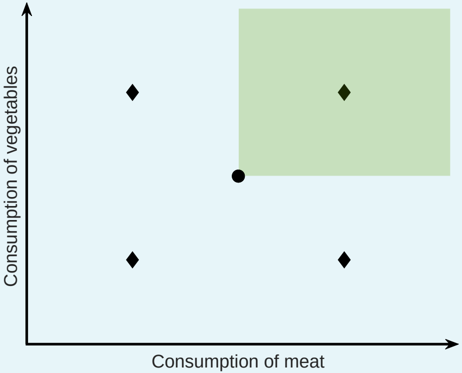
- What about any bundle '**north-east**'?



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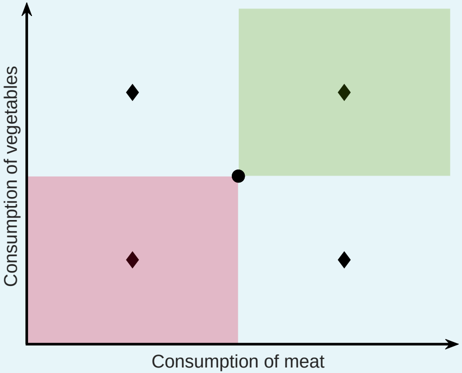

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- Let's compare the alternative options to our original bundle...
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- What about any bundle '**north-east**'?
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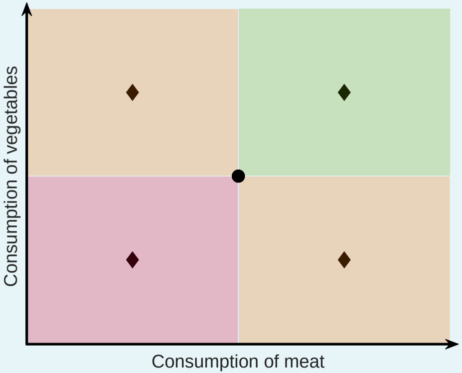
- And bundles '**north-west**' or '**south-east**'?



Introducing the 'utility function'. . .

- Given **non-satiation** (and **completeness**, and **transitivity**), we can represent the consumer's preferences using a **utility function**.

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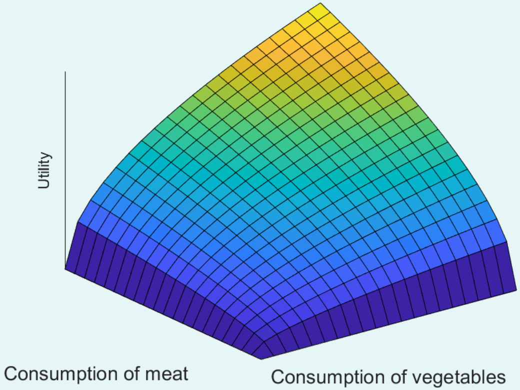

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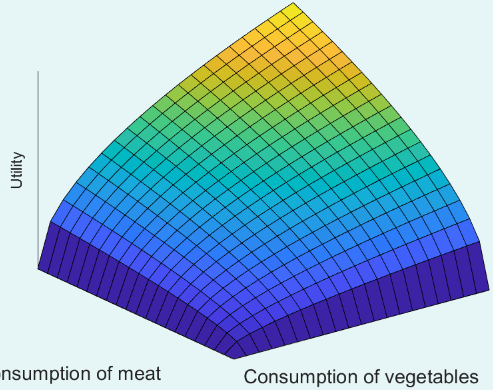
- Given **non-satiation** (and **completeness**, and **transitivity**), we can represent the consumer's preferences using a **utility function**.
- We can add '**contour lines**' to this graph.



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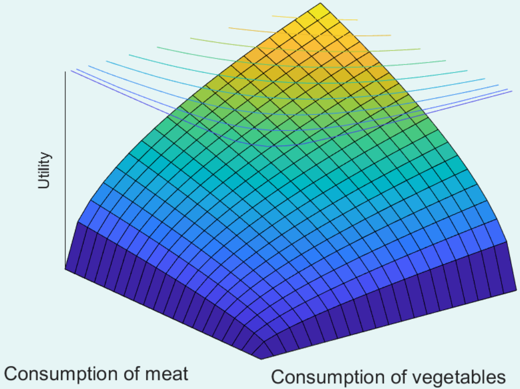
- Given **non-satiation** (and **completeness**, and **transitivity**), we can represent the consumer's preferences using a **utility function**.
- We can add '**contour lines**' to this graph.
- In this context, we can call these **'contour lines'**, **'level sets'**, **'iso-utility curves'**, or **'indifference curves'**.



"I've never seen 'contour lines' before. . ."

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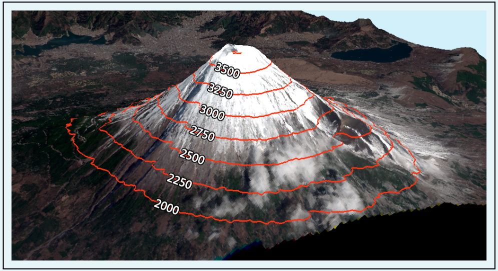

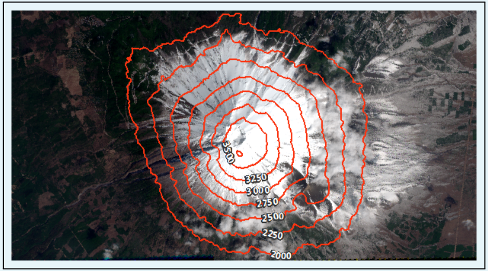

GIS Geography: `https://gisgeography.com/contour-lines-topographic-map/`

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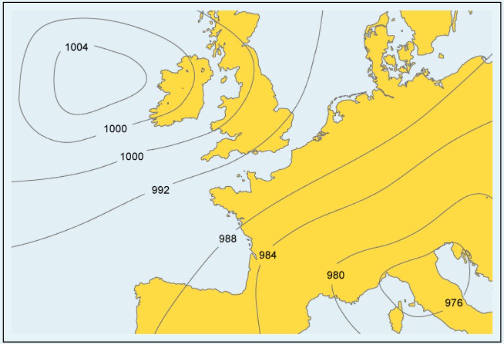
“I’ve never seen ‘contour lines’ before...”



BBC Bitesize: https://www.bbc.co.uk/bitesize/guides/zstcv9q/revision/5

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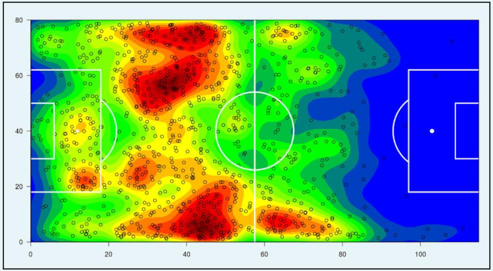
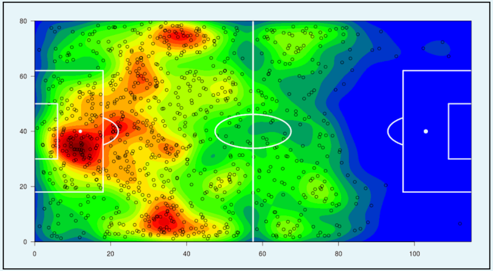
Sporting Kansas City



New York City FC



Sean Steffen:

https://www.americansocceranalysis.com/home/2015/8/17/shot-limiting-bringing-the-heat-maps

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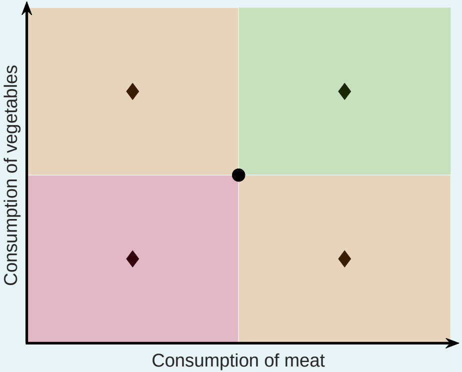
Using **indifference curves** to describe preferences

- What happens if we add indifference curves to our earlier graph?



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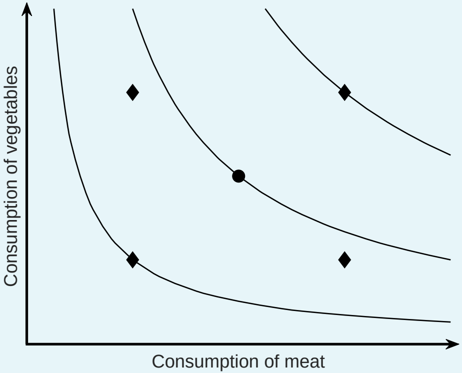

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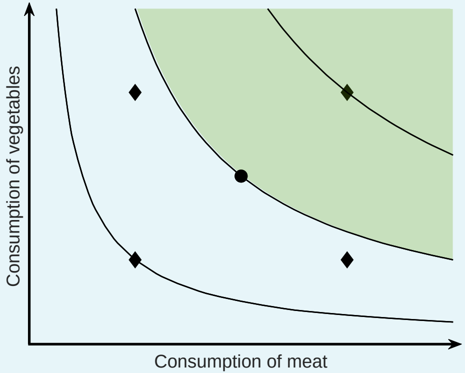

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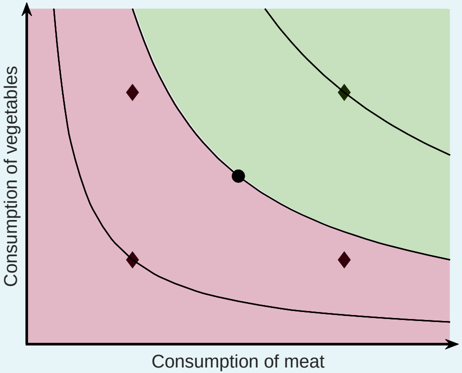
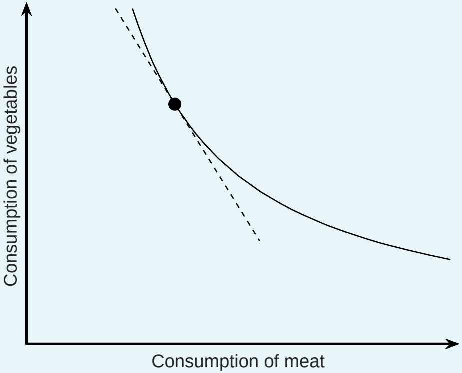


Introducing the **Marginal Rate of Substitution**

- The **slope** of the indifference curve is called the '**Marginal Rate of Substitution**'.



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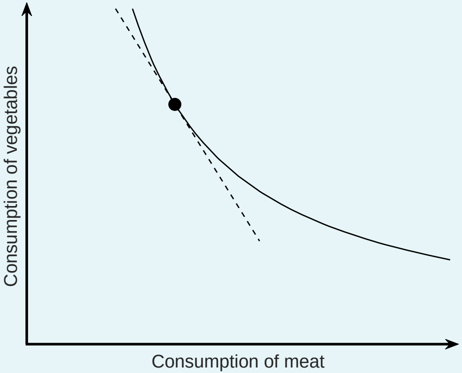

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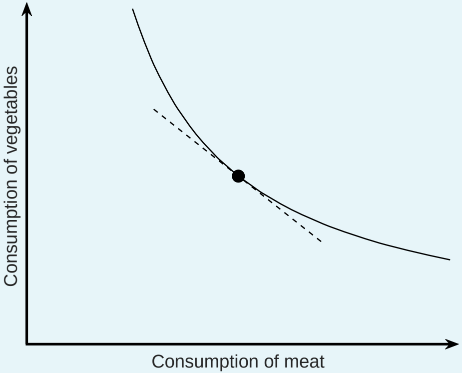

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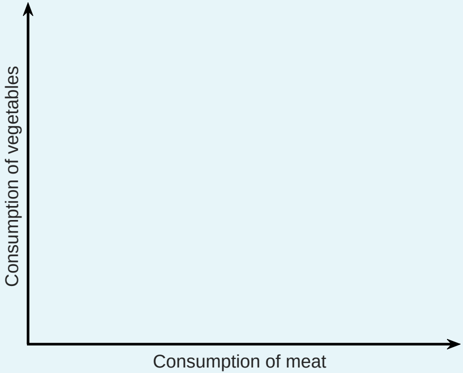

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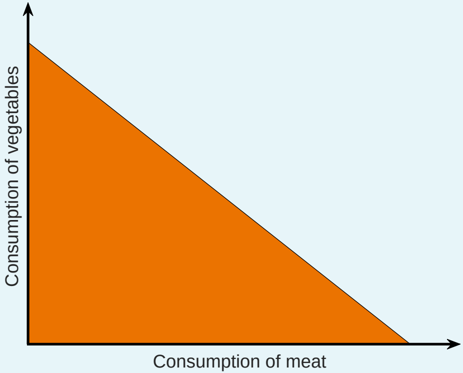

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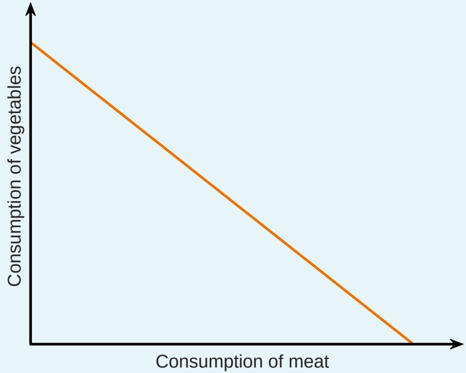

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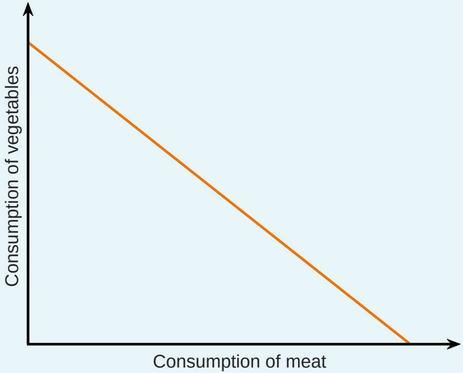

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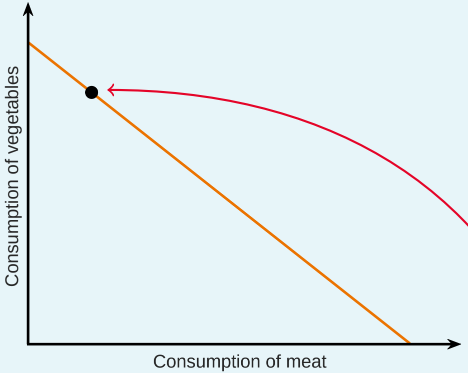

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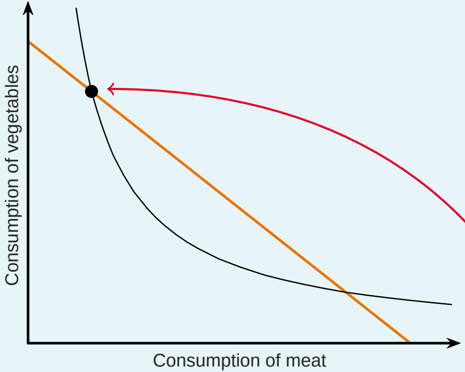

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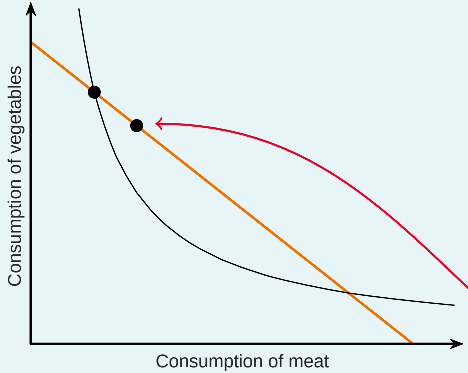

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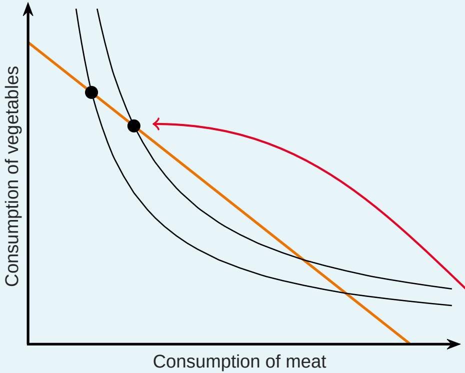

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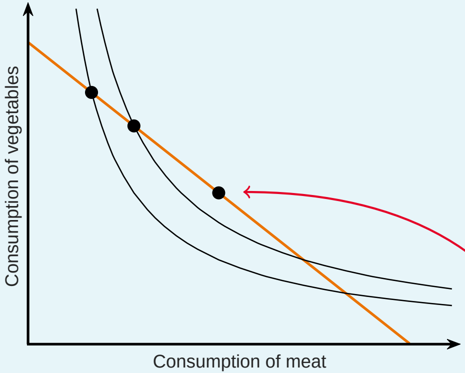

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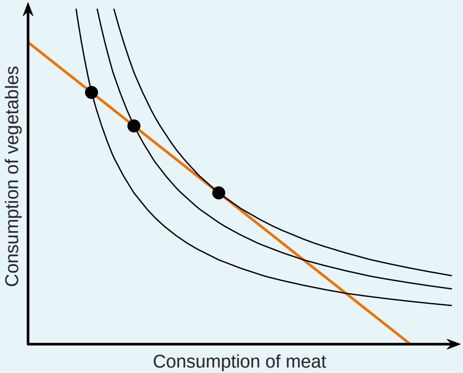

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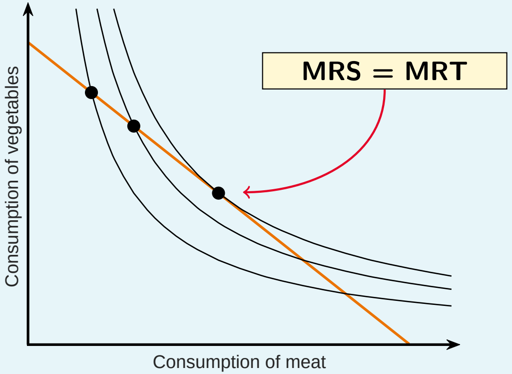

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In the context of an economic model, a **counterfactual** involves changing some model parameter, to see how the model predictions will change.

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We will consider two kinds of counterfactuals:

1. What happens if the **price of meat** changes?

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We will consider two kinds of counterfactuals:

1. What happens if the **price of meat** changes?
2. What happens if **income** changes?

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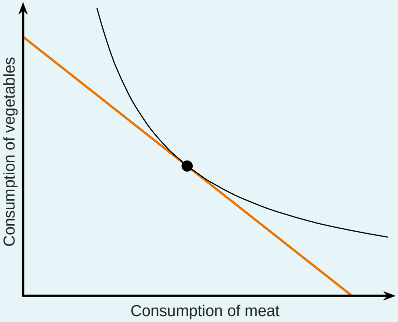
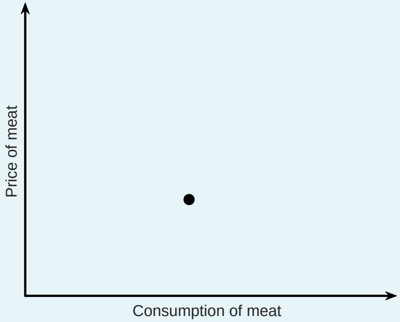
Counterfactual 1: Changing the price of meat





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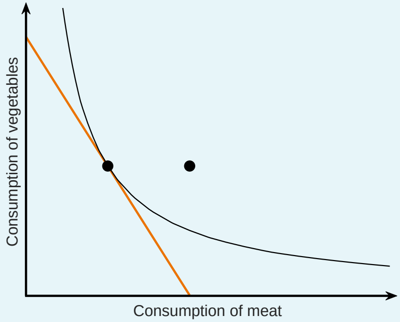

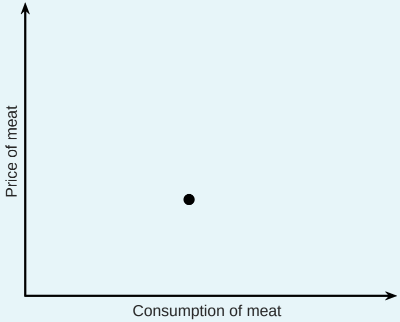

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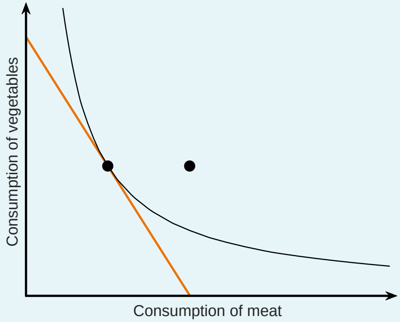

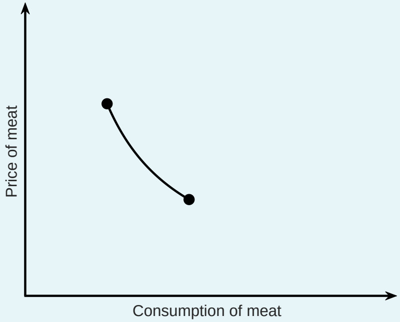

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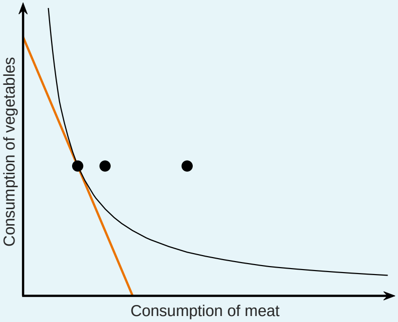

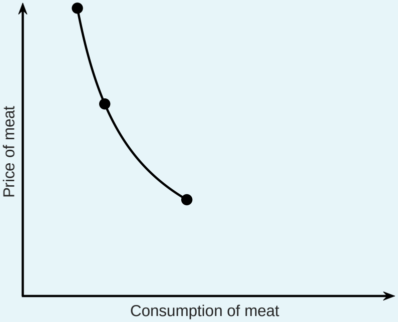

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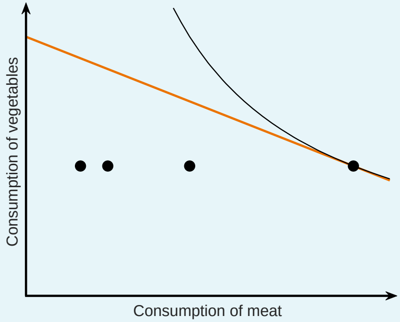

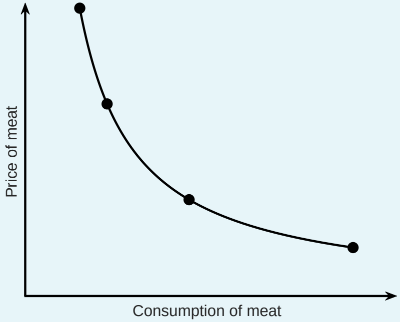

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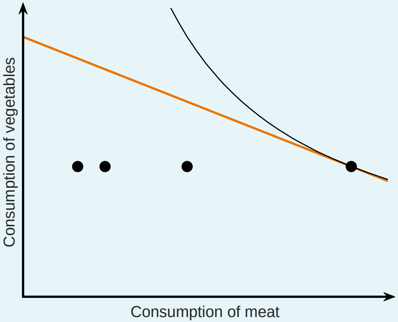
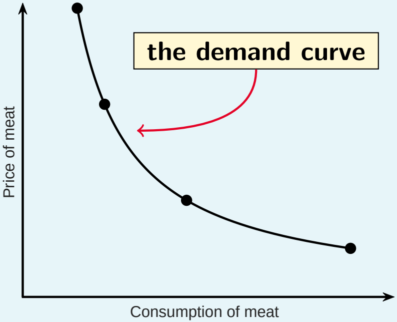
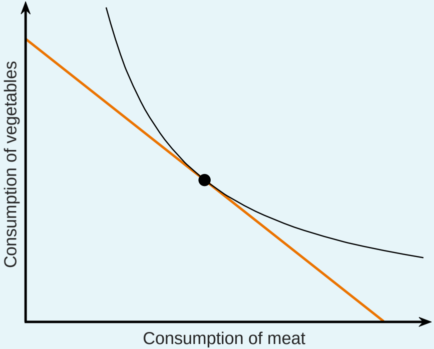




Substitutes and complements



- Notice that the consumption of vegetables did not change. This need not be the case.

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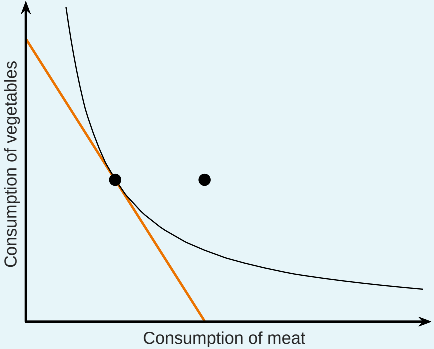

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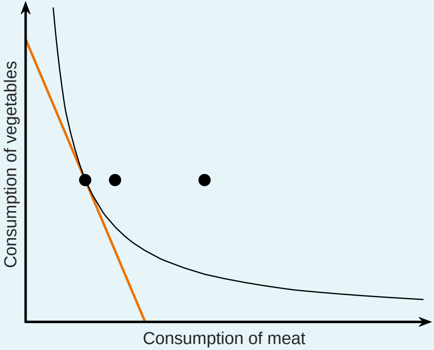

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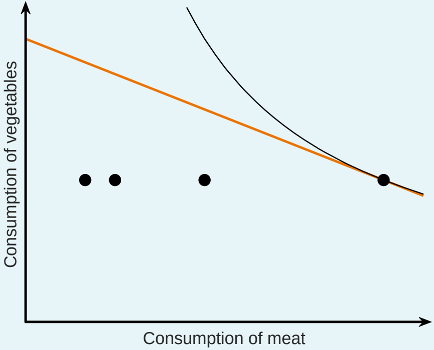

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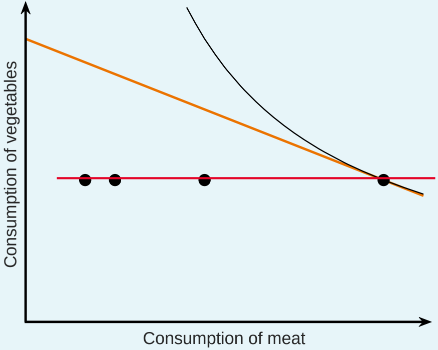

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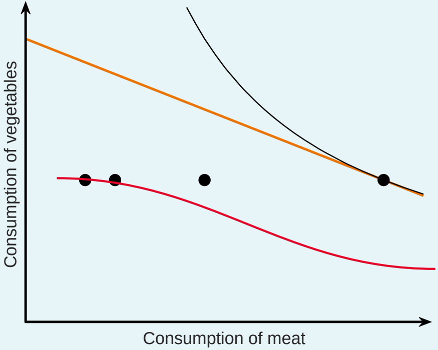
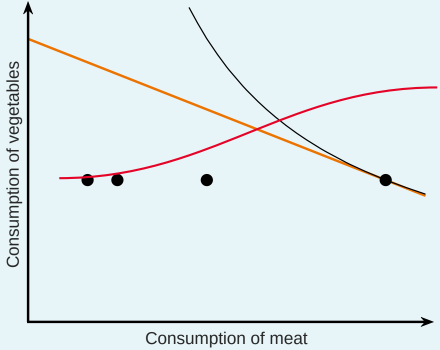


- Notice that the consumption of vegetables did not change. This need not be the case.
- If consumption of vegetables **increases** as the price of meat **increases**, vegetables and meat are **substitutes**.

Substitutes and complements



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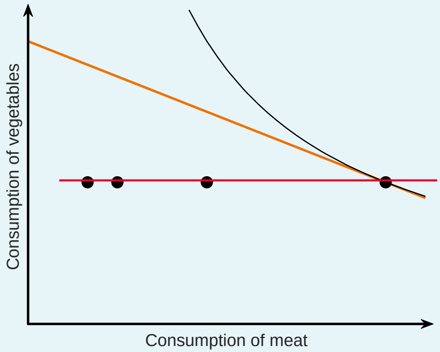

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- If consumption of vegetables **decreases** as the price of meat **increases**, vegetables and meat are **complements**.
- Here, meat and vegetables are **neither** substitutes nor complements.

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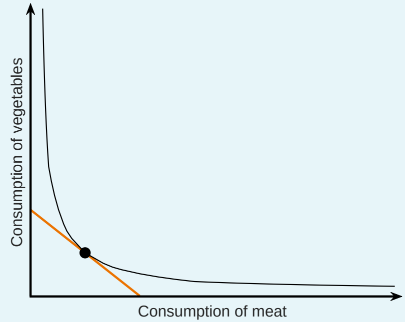
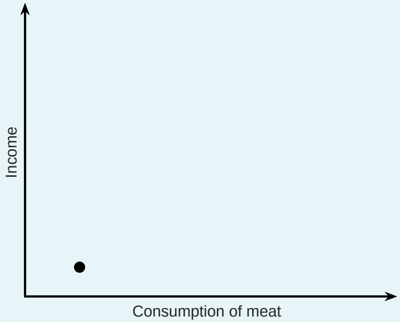
Counterfactual 2: Changing household income





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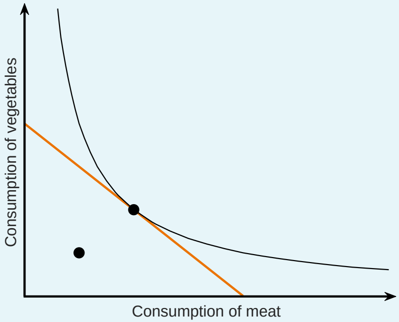

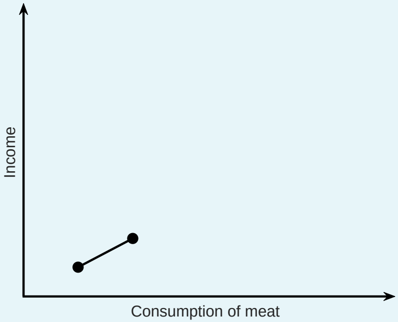

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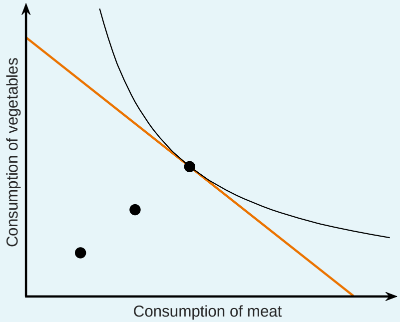

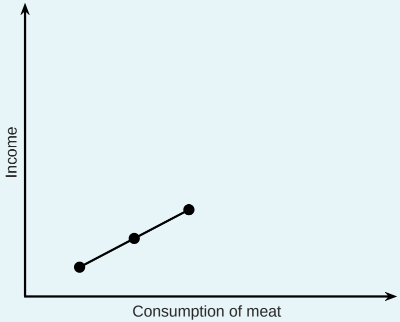

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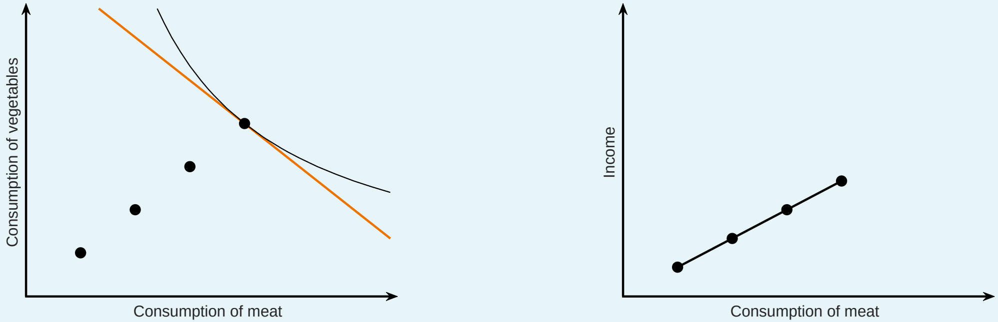

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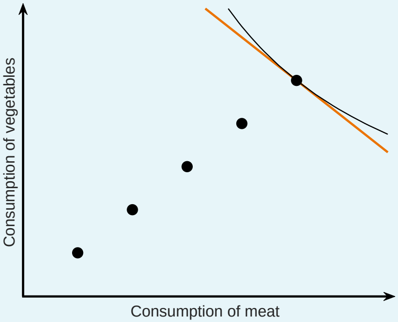

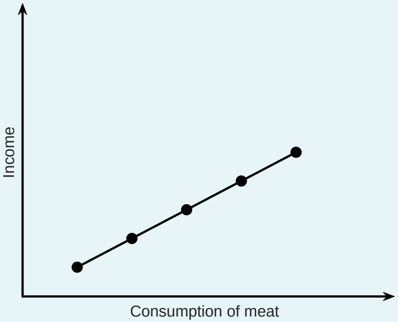

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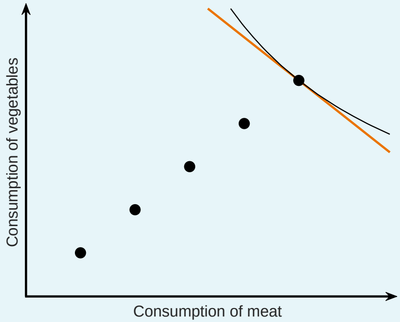

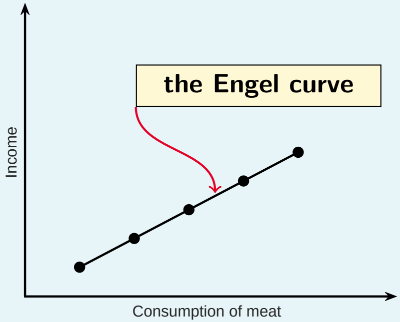

From a stylised household to an ‘ideal policymaker’

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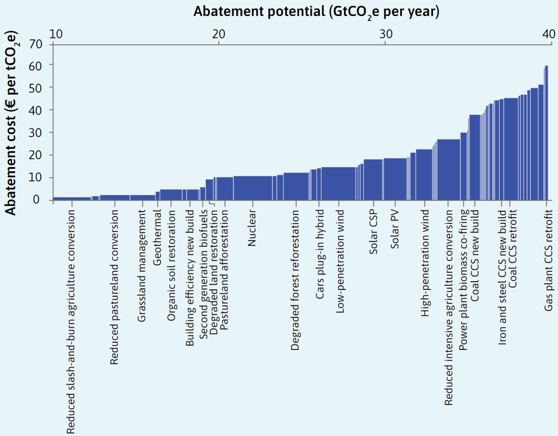
- A question about **effective decision-making**

  How should the '**ideal policymaker**' best serve the citizens' interests?

Ranking policies by abatement cost and potential volume

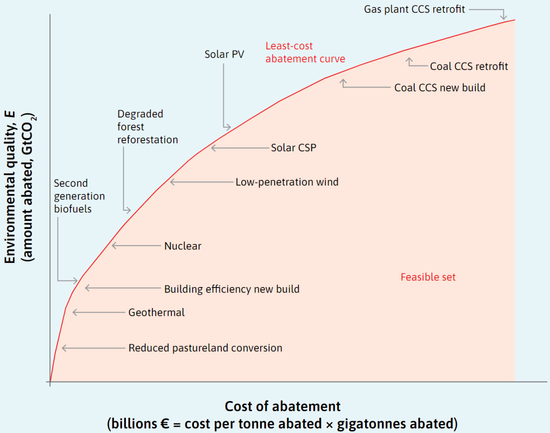
Let's start by ranking potential abatement policies. We rank them by their **abatement cost** and **potential volume**. (*Source:* McKinsey & Company, 2013)



Using the ranking for a ‘**least-cost abatement curve**’

We can then use this ranking to form the ‘**least-cost abatement curve**’.

This shows the trade-off between **total cost of abatement** and **amount of abatement**.



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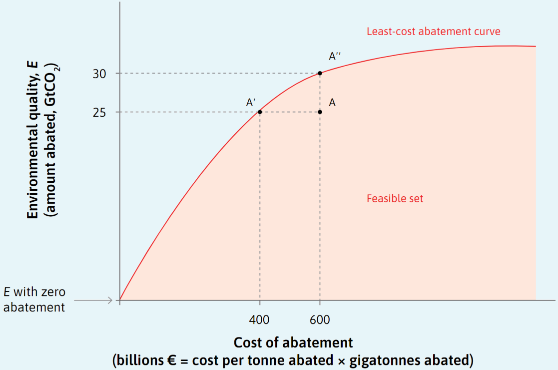
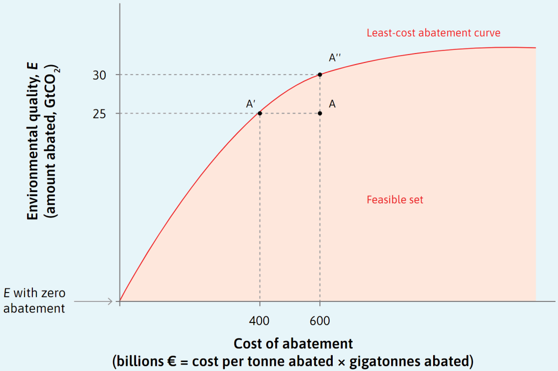


Reversing the x-axis

Now let's **reverse the x-axis**, so that we can visualise in '**consumption and environmental quality space**'.

(**Why** is this useful?)



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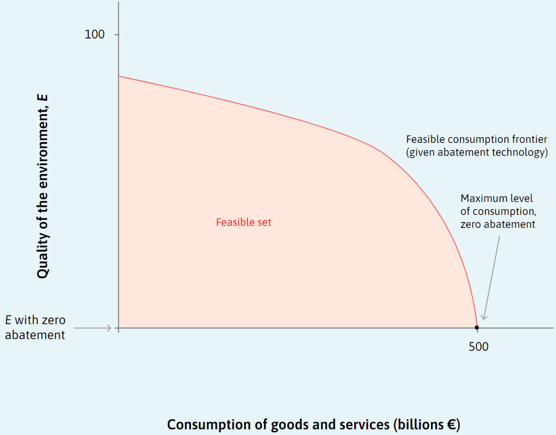

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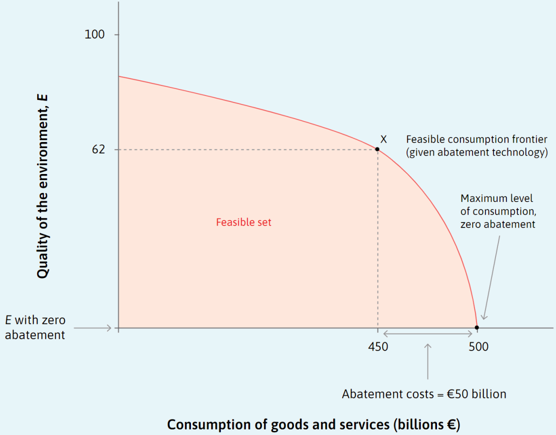
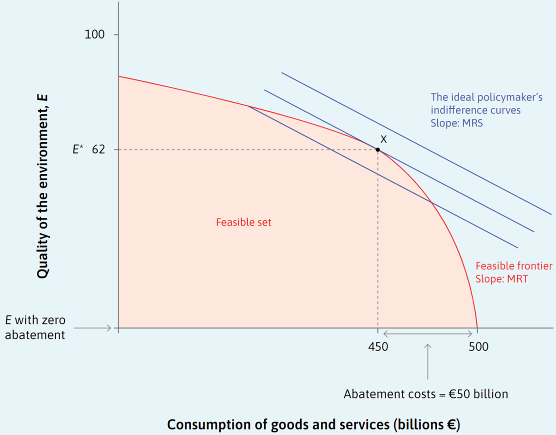


Adding indifference curves

Now let's add **indifference curves** for the policymaker. (For simplicity, we will draw these as straight lines – but they don't need to be.)



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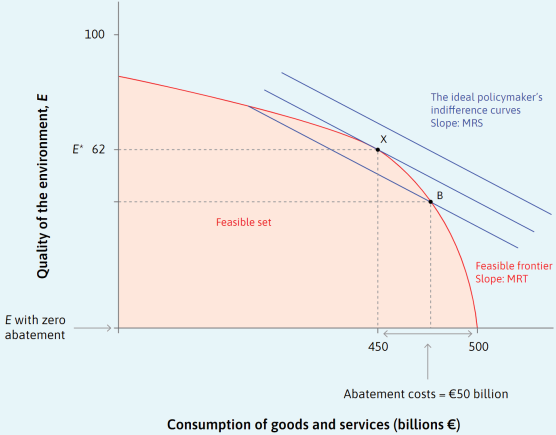

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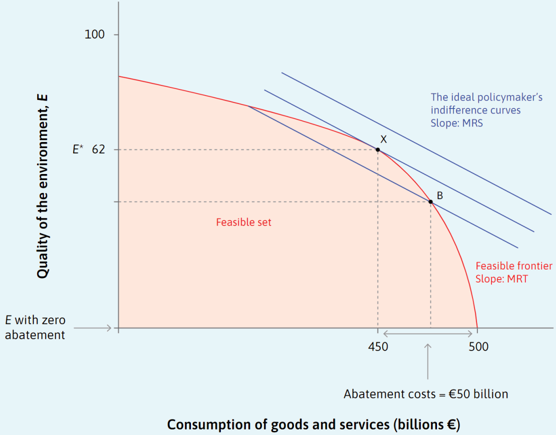
Thought experiment: What would happen if the policymaker were to care less about environmental quality? (Or care more about consumption?)



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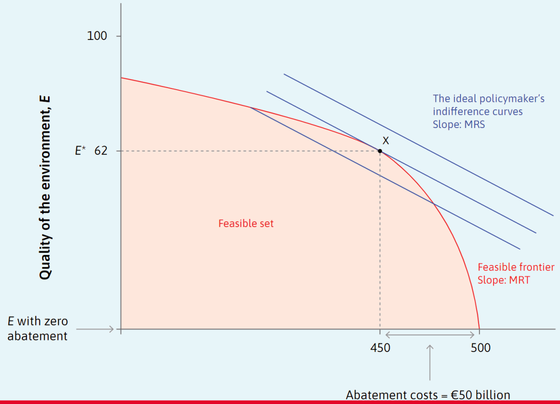
Technological change and carbon abatement

Now suppose that **abatement technologies improve**. How does this change the **feasible set**?



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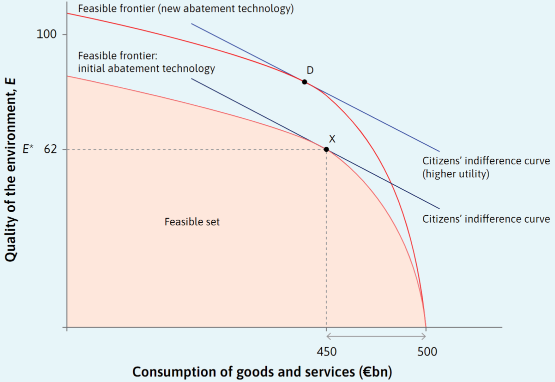
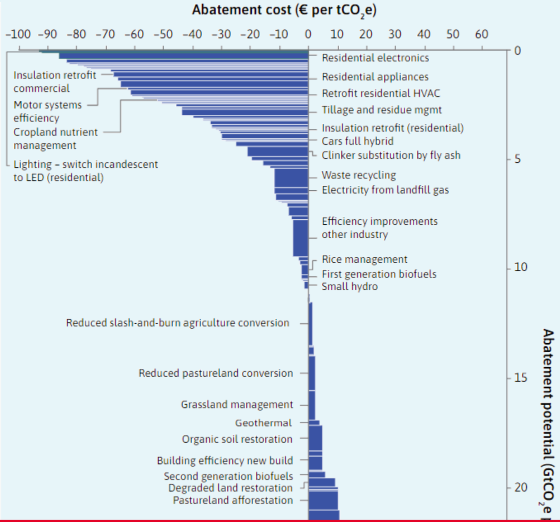
How does this change the **optimal policy** choice? **Why**?



‘Win-win’ policies

Previously, for simplicity, we only discussed measures with a **net cost**.

Fortunately, some abatement measures are ‘**win-win**’ – they have a **net benefit**. Let’s now consider these. . .



‘Win-win’ policies and the optimal policy choice

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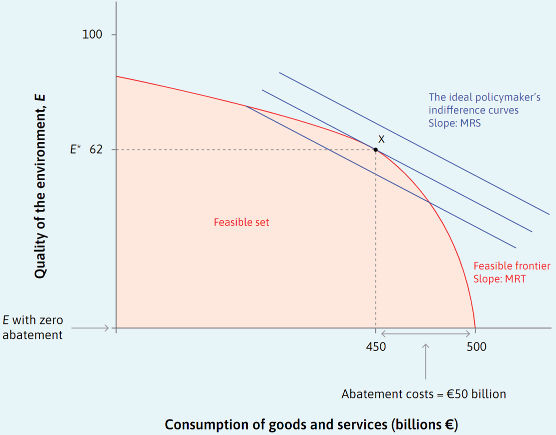

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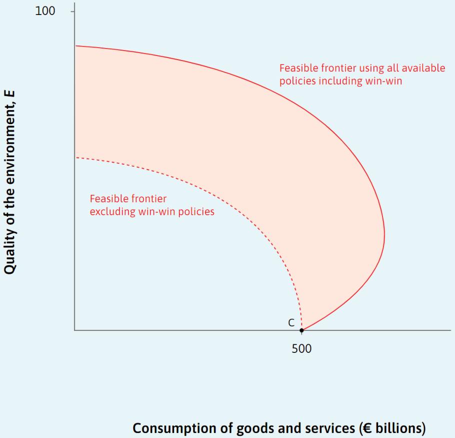
The policies will (i) **push out** the feasible frontier, and (ii) initially, give the feasible frontier a **positive slope**.



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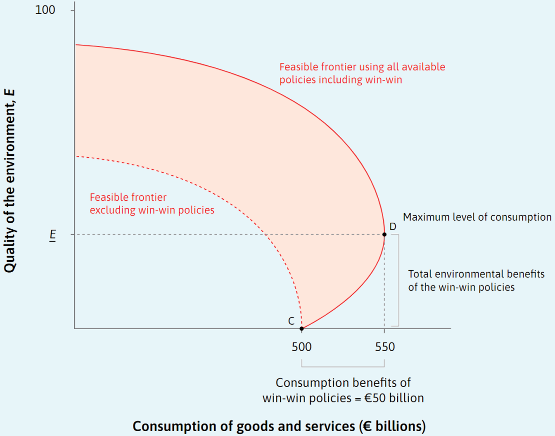
How will this affect the optimal policy choice? (Start by considering a policymaker who cares **only** about consumption. . .)



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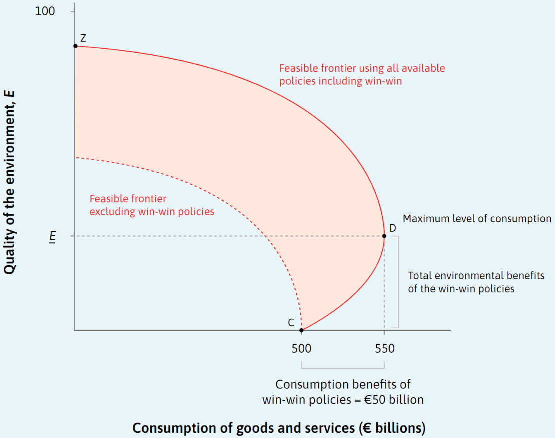
Why, then, might an economy **not** adopt all the 'win-win' policies?



Summary of consumer behaviour

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In summary,

1. We can model consumer behaviour using a **utility function**, and we can represent this using **contour lines** ('**indifference curves**');
2. The shape of the utility function determines how demand responds to changes in prices ('**the demand curve**') and changes in the budget constraint ('**the Engel curve**').
3. We can use the same framework to think about an '**ideal policymaker**', deciding the **optimal level of abatement and optimal mix of abatement policies**.

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Lecture 7: Tuesday 30 September2024

Consumer Preferences

Business Economics, Climate Change and the Environment

Simon Quinn